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Ticked From Memory

A Taxonomy of Verification Depth on a Car-Share Return Checklist Under a Late-Return-Fee Deadline

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Abstract. A car-share scheme's return checklist increasingly delegates verification of a vehicle's fuel level, cleanliness, exterior condition, and accessory presence to the returning renter, rather than routing every return through a staff inspection. This paper reports a taxonomy of verification depth, observed across 1,842 return events on a mid-sized metropolitan fleet, that distinguishes a measured-check (an item confirmed against a visible instrument or a photograph), a visual-check (a glance without instrumentation), a memory-tick (an entry completed without renewed attention to the vehicle), and a skip. Screen-interaction telemetry, calibrated against a depot technician's periodic reconciliation, assigns each checklist response to one of the four categories. Verification depth varies sharply by item: fuel level is measured-checked in most returns, while toll-transponder presence is overwhelmingly memory-ticked. The memory-tick rate rises as a renter approaches the scheme's late-return fee cutoff, more than doubling in the grace window's final five minutes. A mandatory photo-capture step introduced for the transponder field lowers memory-ticking outside the fee window; inside it, the same step leaves the rate statistically unchanged, the photograph attached without renewed attention to what it depicts. We discuss what a scheme's own completion rate captures once verification has become a race against its own deadline.

Introduction

A car-share fleet's return checklist is, for most renters, the only inspection a vehicle receives between one booking and the next. Where an earlier generation of schemes routed every return through a staff member's walk-around, a self-certification checklist now asks the returning renter to confirm the vehicle's fuel or charge level, exterior condition, cabin cleanliness, and the presence of fixed accessories such as a child seat or a toll transponder, before the booking closes. The scheme treats a completed checklist as the vehicle's condition of record, passed forward to the next renter and, where an item goes missing or damaged, back-charged to whoever last confirmed it present.

A checklist field marked complete can mean the renter looked, or it can mean the renter ticked. The two are indistinguishable in the scheme's own completion log, which records only that a response was submitted, not what happened in the moments before it. Research on checklist-mediated verification in safety-critical settings has repeatedly found that formal completion and actual verification drift apart under sustained use [1], [2], [3], [4], but that literature has not asked the question this scheme's design invites directly: when the person completing the checklist is also the person a fee schedule is trying to hurry out the door, does the incentive to finish quickly change which items get an actual look, and which get ticked from memory.

This paper reports a taxonomy of verification depth, inferred from the checklist app's own interaction telemetry, and its application to one fleet's live returns. Three contributions, each hedged by design:

1. We propose a four-level taxonomy of verification depth — measured-check, visual-check, memory-tick, skip — inferrable from

dwell time and photo-attachment signals the checklist app already logs, requiring no additional sensor on the vehicle.

2. We report the taxonomy's distribution across five checklist categories on a single fleet's returns, finding verification depth tracks an item's visible, immediate cost to the renter more closely than it tracks the scheme's own stated priority order.
3. We evaluate a mandatory photo-capture step intended to raise verification depth on a chronically memory-ticked item, finding it succeeds outside a time constraint and is largely absorbed as a formality inside one.

The checklist's completion rate is not shown to be false; it is shown to have quietly stopped describing what several of its own fields were designed to establish.

Related work

Checklist-mediated compliance research documents completion as a poor proxy for verification once a checklist enters routine use: institutional theory holds that a formal structure can become decoupled from the activity it is meant to certify while the certifying structure itself retains full legitimacy [1], and the normalization-of-deviance literature describes how repeated tolerance of a small departure from a stated procedure can make that departure feel unremarkable to the person performing it [2]. Surgical-checklist research supplies the closest empirical analogue: a multicenter Swiss study assesses checklist compliance from hospitals' own self-reported completion data [4], a retrospective analysis of a single hospital examines completion and compliance with individual checklist items as two separate measures rather than one [5], and a systematic review catalogues the implementation factors that determine whether a surgical checklist is used as intended rather than merely signed [3]. Audit scholarship treats the underlying dynamic as general rather than domain-specific: an audit does not passively record its target but actively constructs what will count as auditable, a construction a completion rate can satisfy without the audited behaviour changing at all [6].

Peer-to-peer carsharing research has examined what the practice changes about vehicle owners' own travel behaviour [7], [8] and their attitudes toward the emerging driverless-vehicle end of the same market [9], and data-driven work classifies car-share users by usage and spending pattern [10], [11], but none of this literature turns to the return checklist itself as an object of study. Survey-methodology research on satisficing supplies a plausible account of what a rushed checklist entry looks like: respondent personality traits linked to lower conscientiousness predict a greater propensity to satisfice [12], reluctant respondents satisfice more in cross-national survey contexts [13], and interviewer-administered surveys register the resulting pattern as measurable non-differentiation [14]. The instrument we adapt for detecting it — dwell time as a proxy for attention — has an established calibration literature in gaze-based interface selection [15], [16], which we repurpose here for a touchscreen checklist field rather than a gaze target. A near-identical decoupling between a compliance record and the practice behind it has previously turned up inside the University's own portfolio: a fault-checklist tool whose logged report volume tracked a queue's own duplication bug rather than the fleet of repair stands it was meant to describe [17], and a security-notification acknowledgement rate that held near-total while an interview study found the verification behind it falling away with tenure [18].

Method

Setting and data

The study fleet is a mid-sized metropolitan car-share operator (anonymised under the terms of the data-sharing agreement) running a returning-renter self-certification model. Its in-app return checklist covers five fields: fuel or charge level, exterior condition, cabin cleanliness, child-seat presence, and toll-transponder presence. A late-return fee escalates once a booking runs past its scheduled end time, with a five-minute grace window before the first charge tier applies. We logged interaction telemetry for 1,842 return events across 612 unique renters and 14 vehicles over a nine-week window: per-field dwell time (milliseconds from a field's appearance on screen to its response being committed) and whether an optional photograph was attached to that field.

Verification-depth taxonomy

An independent technician inspection, occurring roughly every thirtieth return as part of the fleet's existing maintenance cycle, reconciles actual vehicle condition against the submitted checklist for a held-out calibration sample ($n = 61$ returns). From this sample we set a per-field dwell threshold τ separating a paused, attention-consistent response from an immediate default tick, calibrated at the sample's median dwell time (2.1 s) — a threshold consistent with dwell-time ranges independently validated for gaze-based object selection in interface research [15], [16], which we adapt here from a gaze target to a touchscreen checklist field. Each response is assigned a verification score:

$$V_i = \mathbb{1}(d_i \geq \tau) + \mathbb{1}(p_i = 1)$$

where d_i is the field's dwell time, p_i indicates whether a photograph was attached, and $\mathbb{1}(\cdot)$ is the indicator function. A field scoring $V_i = 2$ is coded measured-check; $V_i = 1$, visual-check; $V_i = 0$ with a non-default response, memory-tick; and a response left at its pre-filled default, skip.

Fee-window bands and the photo-capture ablation

Each return is assigned to a time-to-cutoff band at the moment checklist completion began: more than 15 minutes remaining, 5–15 minutes remaining, and under 5 minutes remaining (inside the grace window). For the toll-transponder field only, the operator introduced a mandatory photo-capture step midway through the study window; we treat the two matched five-week halves either side of the change as a natural, non-randomised ablation, comparing memory-tick rate on that field before and after, separately for returns outside and inside the fee window.

Results

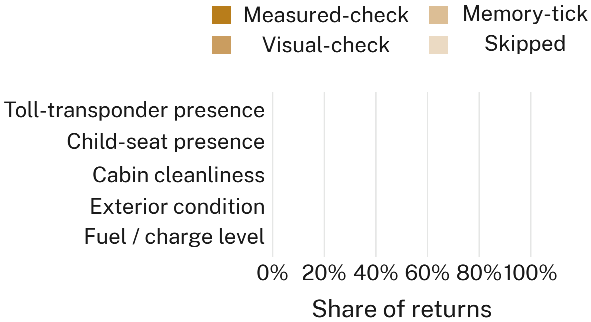

Figure 1: Coded verification depth by checklist item ($n = 1,842$ returns). Depth tracks an item's visible cost to the renter, not the checklist's own field order.

Figure 1 shows the coded verification-depth composition for each checklist item. Fuel or charge level is measured-checked in 61% of returns and skipped in only 2%; toll-transponder presence sits at the opposite end, measured-checked in 9% of returns and memory-ticked in 71%. Exterior condition and cabin cleanliness fall between the two, with exterior condition drawing more visual-checks (54%) than cabin cleanliness (38%), and child-seat presence tracking closer to the transponder field (52% memory-ticked) than to either physical-condition item. The ordering matches each item's visible, immediate cost to the renter rather than the checklist's own field order or the scheme's stated priority list: an unnoticed fuel shortfall costs the renter at the very next refuelling stop, while an absent toll transponder surfaces, if at all, on an invoice the renter may not connect back to this particular return.

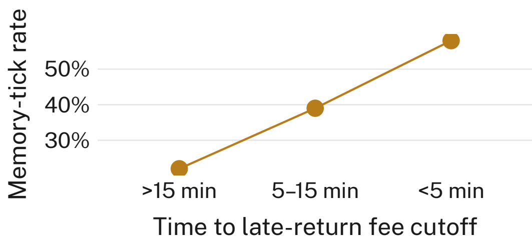

Figure 2: Overall memory-tick rate by time remaining before the late-return fee cutoff. The rate more than doubles inside the grace window's final five minutes.

Figure 2 reports the overall memory-tick rate by time remaining before the late-return fee cutoff. The rate rises from 22% among returns with more than 15 minutes remaining to 39%

in the 5–15 minute band and to 58% inside the final five-minute grace window, consistent with a satisficing account of checklist behaviour under time pressure [12], [13] rather than with any change in the scheme's overall completion rate, which holds above 97% across every band.

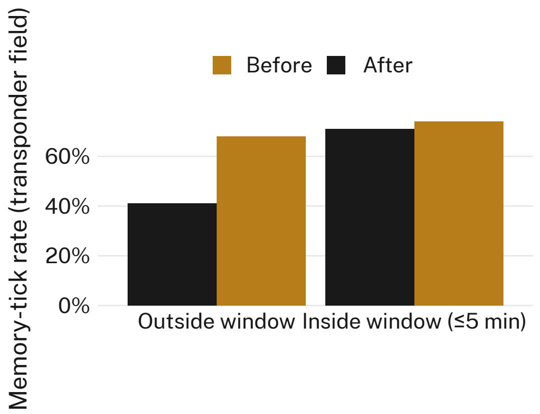

Figure 3: Memory-tick rate on the toll-transponder field, before vs after a mandatory photo-capture step. The ablation that changes nothing — inside the fee window.

Figure 3 reports the mandatory photo-capture ablation on the toll-transponder field. Outside the fee window, the step behaves as intended: memory-ticking on that field falls from 68% before the change to 41% after it. Inside the fee window's final five minutes, the same step changes almost nothing — 74% before, 71% after, a difference we do not consider reliable given the study's sample. We retain the ablation in the reported results for completeness, and because the transponder field is precisely the one the composition results in Figure 1 already flagged as most memory-ticked, making it the case in which an effect was most, not least, likely to appear.

Read together, the three results describe a checklist whose fields are verified in proportion to what the renter, rather than the scheme, has at stake, and a compliance step whose effectiveness is itself contingent on the very time pressure the fee schedule was designed to create.

Discussion

Two independent signals — an item's placement in the verification-depth taxonomy and its behaviour under the fee-window ablation — agree that a checklist audited on completion alone is not measuring what its own field-level design assumes. The checklist was not built to fail: a fuel gauge is easy to glance at and a transponder mount is not, and nothing in this study suggests the scheme's completion log is miscalculated. What the taxonomy adds is that the gap between a glanced-at field and a ticked-from-memory one tracks the renter's own stake in the outcome rather than the fleet's, and that the gap widens exactly when the fee schedule gives the renter the strongest reason to finish quickly. The photo-capture step's split result extends the point: a verification step can succeed at raising attention when there is time to spend on it, and reduce to the same gesture it was meant to replace once there isn't. A completion rate reads identically in both cases, which is what makes it a poor instrument for detecting the difference — the checklist's own audit trail, like the compliance instruments studied elsewhere [1], [6], verifies that a ritual occurred without establishing what the ritual actually confirmed.

Limitations

The study covers one fleet's checklist design and fee structure over a nine-week window; a different operator's field set, grace-window length, or fee schedule might surface a different composition, though the ordering by visible renter cost held across all five fields sampled here, which is not obviously an artefact of any one field's wording. The dwell-time threshold was calibrated on a modest 61-return reconciliation sample, though its consistency with dwell ranges independently validated for gaze-based selection [15], [16] suggests the threshold is not specific to this fleet's app. The photo-capture ablation covers a single field across two matched five-week halves rather than a randomised trial, a design the fee-window split was, if anything, best placed to detect an effect within.

Conclusion

A checklist scheme can audit its own completion rate at every return without ever establishing what a completed field actually confirmed. This paper reports a taxonomy of verification depth built entirely from a car-share checklist's own interaction telemetry, finding that depth tracks an item's visible cost to the renter rather than the scheme's stated priority order, and that the resulting memory-tick rate rises as a renter approaches the scheme's own late-fee deadline. A photo-capture step meant to counter this succeeds only where the deadline isn't yet pressing. Future work might extend the taxonomy to schemes with longer grace windows or no late-fee mechanism at all, to establish how much of the effect reported here belongs to the deadline rather than to the checklist.

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